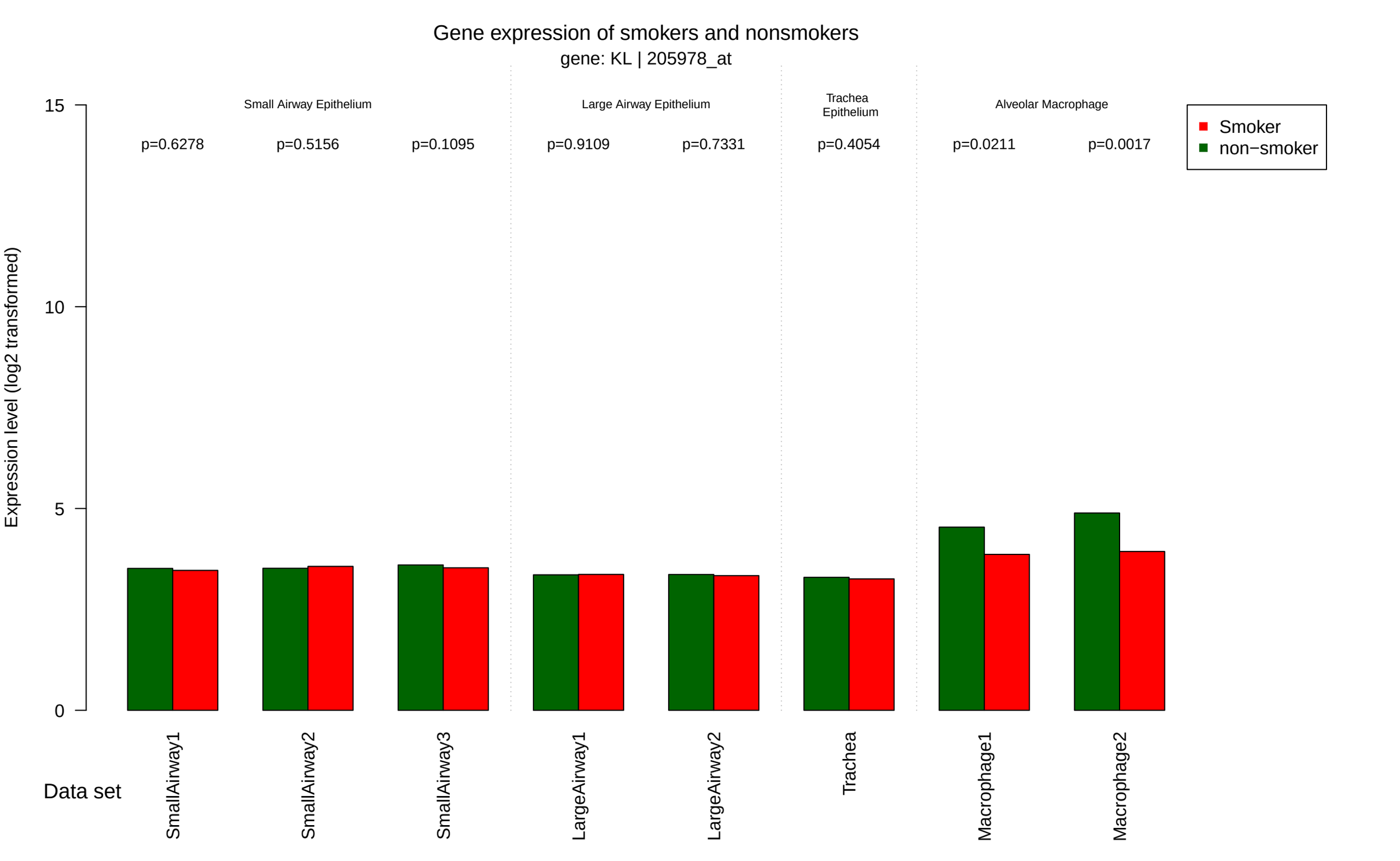
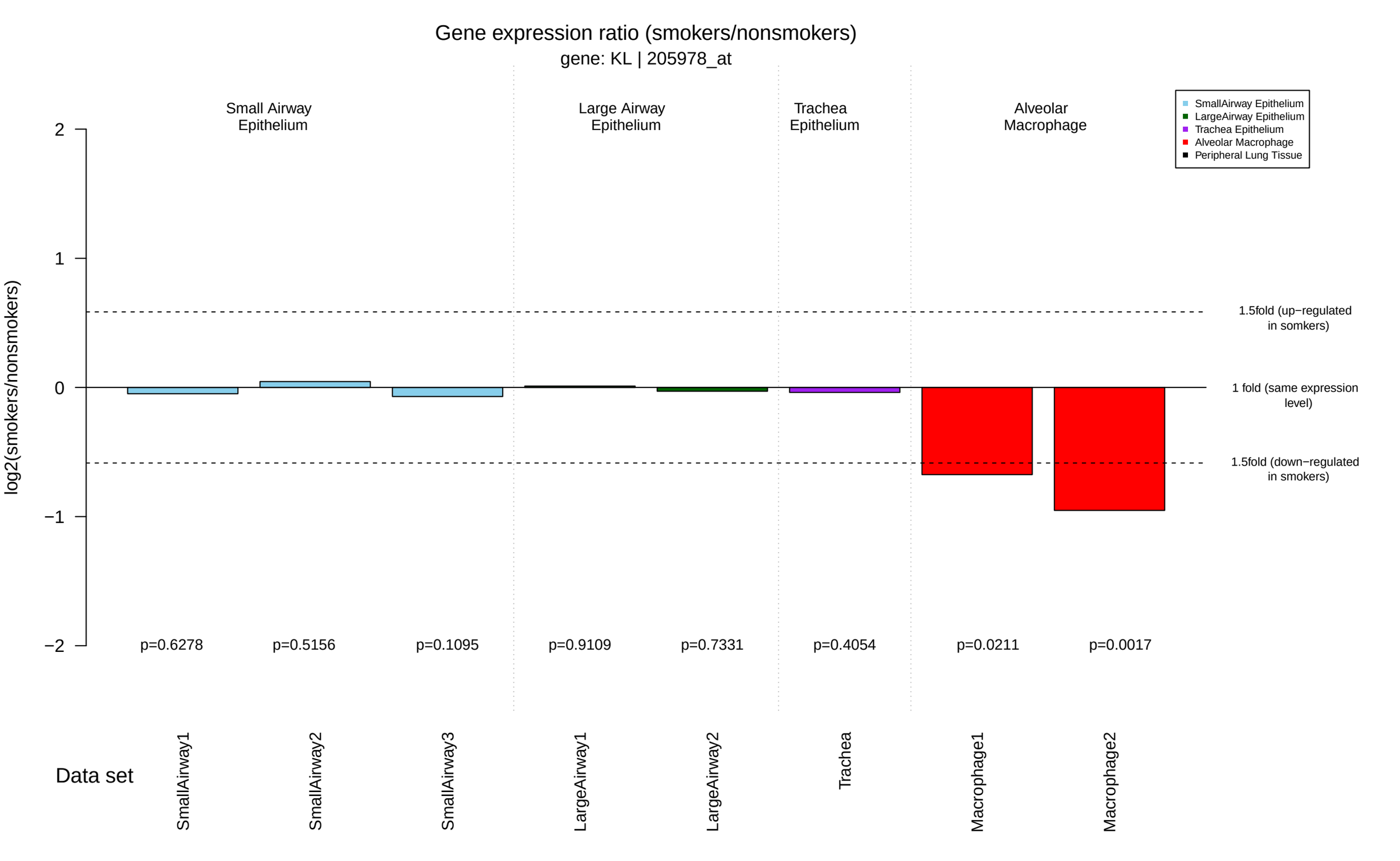
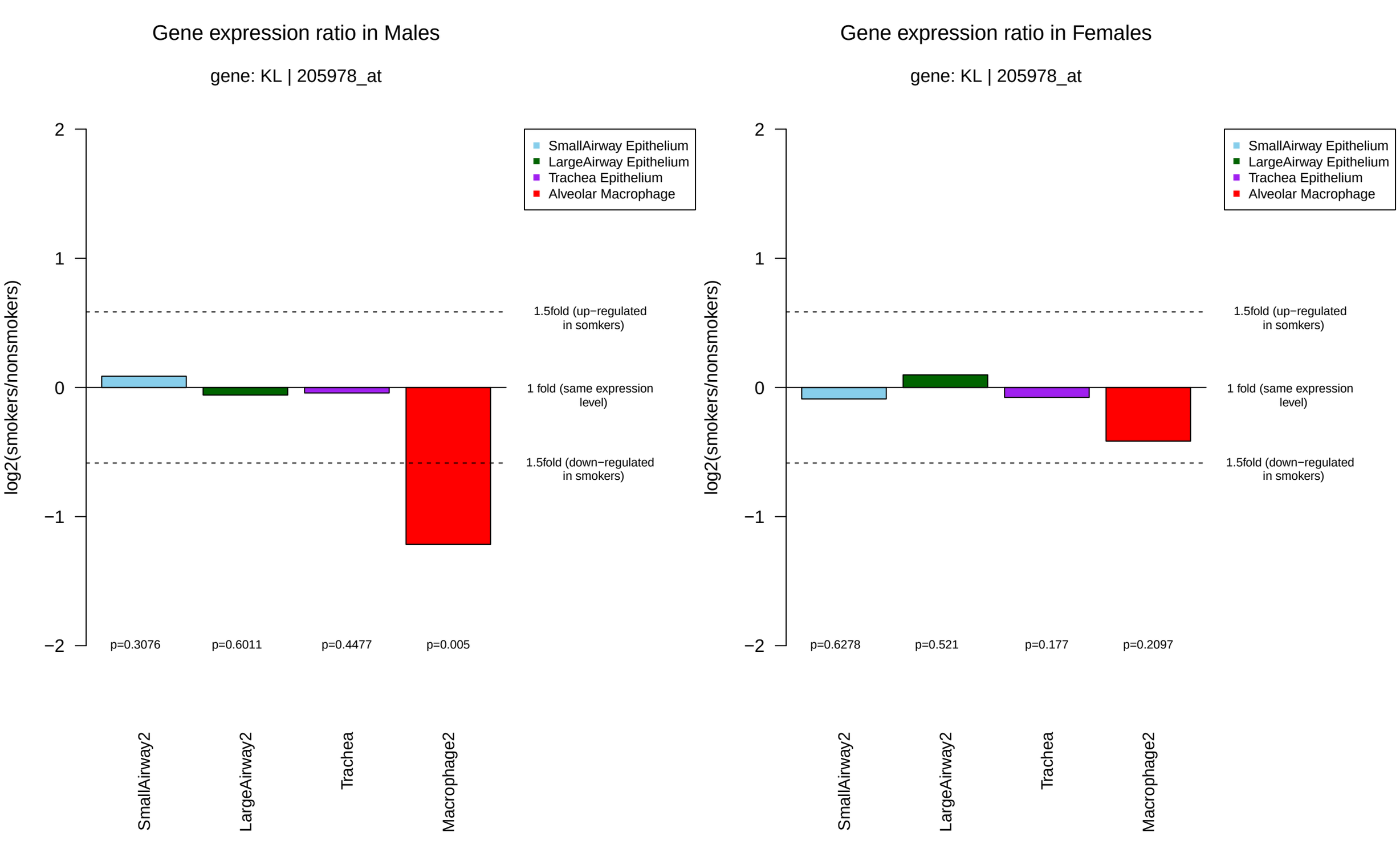
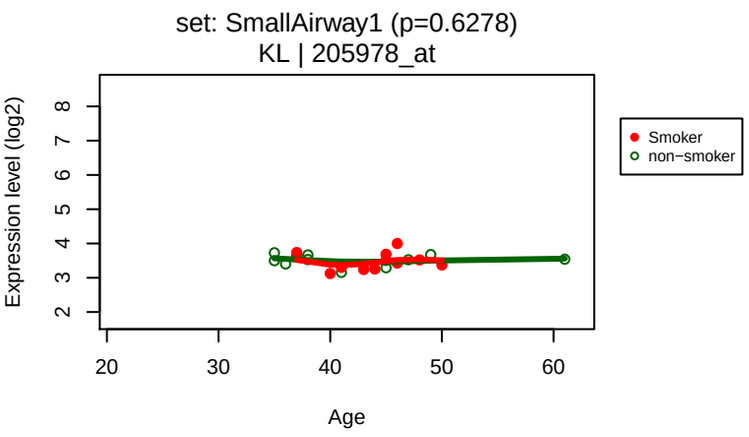
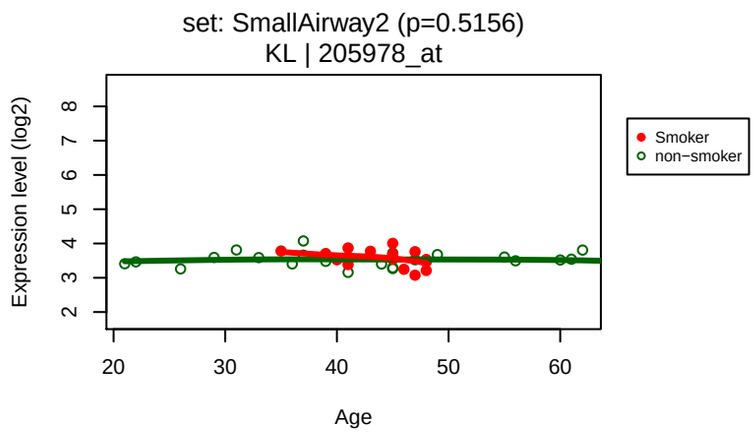
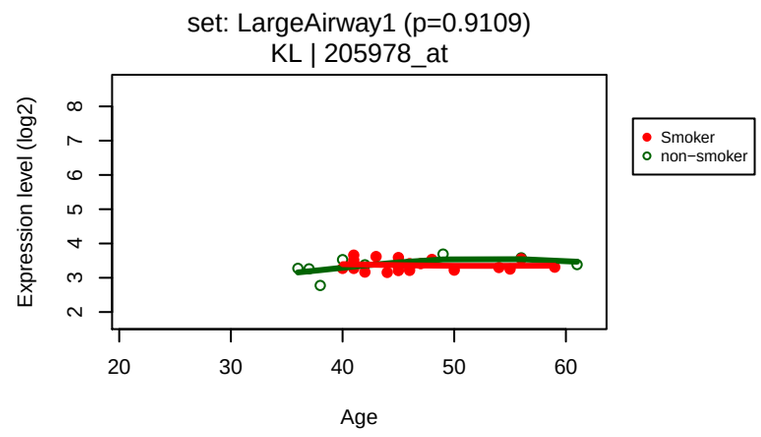

Gene expression of smokers and nonsmokers
gene: KL | 205978_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6278
p=0.5156
p=0.1095
p=0.9109
p=0.7331
p=0.4054
p=0.0211
p=0.0017
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

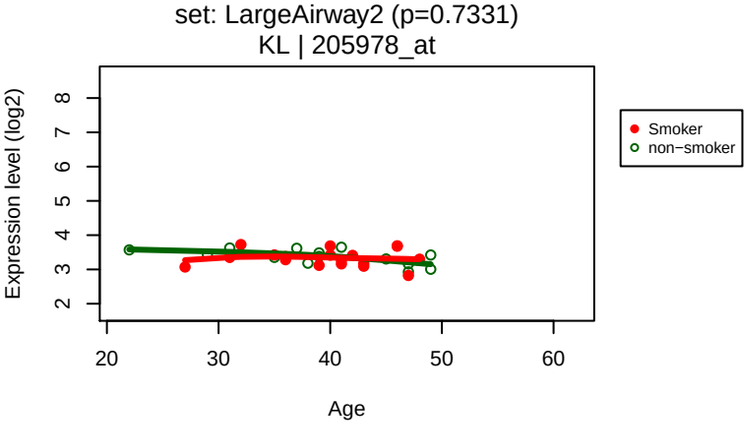

Gene expression ratio (smokers/nonsmokers)
gene: KL | 205978_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6278
p=0.5156
p=0.1095
p=0.9109
p=0.7331
p=0.4054
p=0.0211
p=0.0017
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

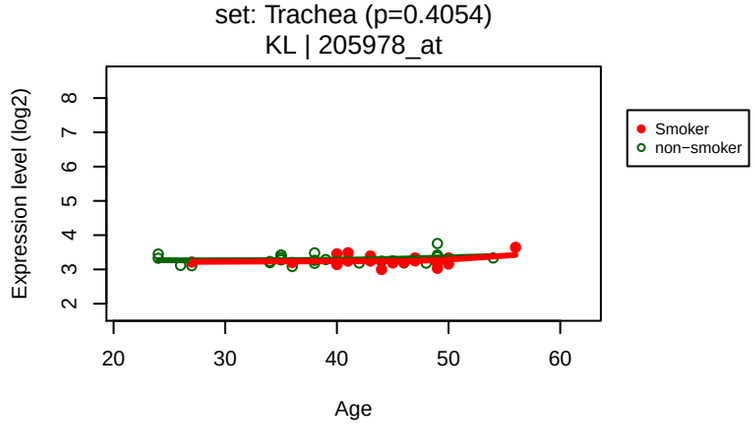

Gene expression ratio in Males
gene: KL | 205978_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3076
p=0.6011
p=0.4477
p=0.005
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KL | 205978_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6278
p=0.521
p=0.177
p=0.2097
SmallAirway2
LargeAirway2
Trachea
Macrophage2

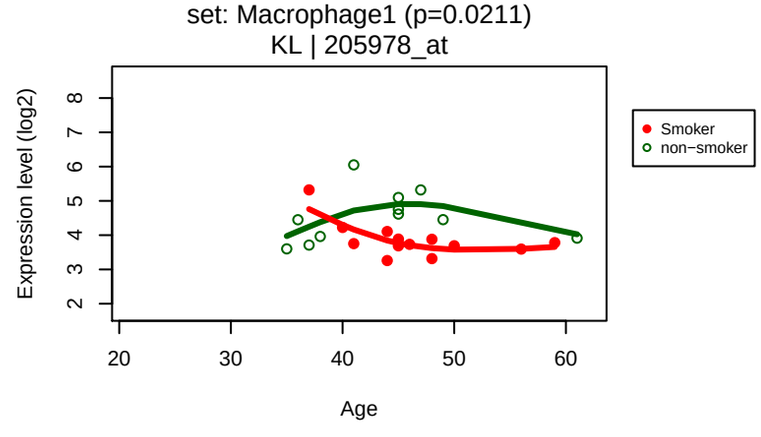

set: SmallAirway1 (p=0.6278)
KL | 205978_at
Expression level (log2)
Age
Smoker
non−smoker

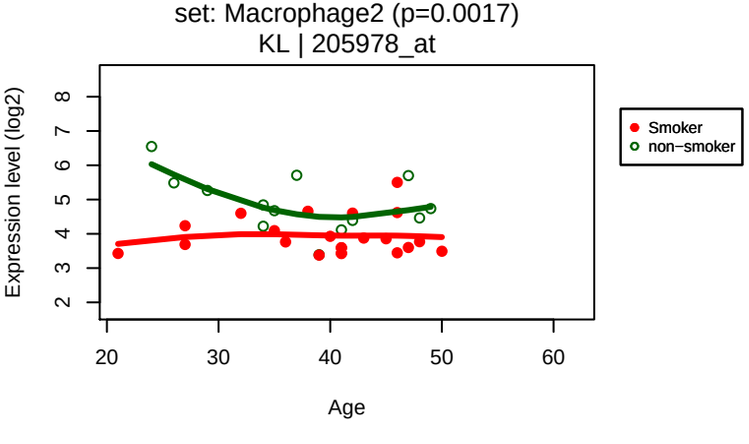

set: SmallAirway2 (p=0.5156)
KL | 205978_at
Expression level (log2)
Age
Smoker
non−smoker

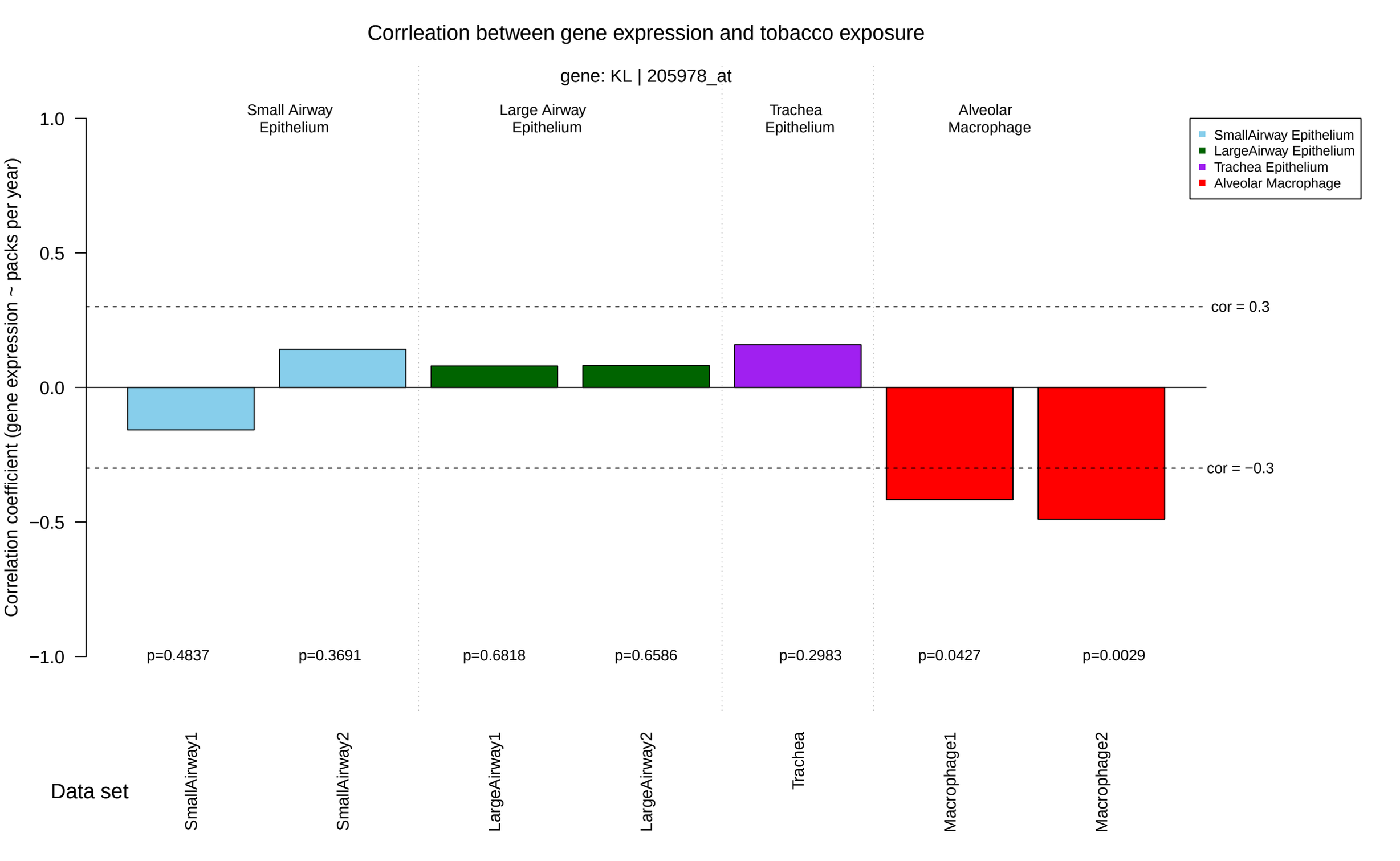

set: LargeAirway1 (p=0.9109)
KL | 205978_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7331)
KL | 205978_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4054)
KL | 205978_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0211)
KL | 205978_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0017)
KL | 205978_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KL | 205978_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4837
p=0.3691
p=0.6818
p=0.6586
p=0.2983
p=0.0427
p=0.0029
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2